

EXCHANGE RATE ANALYSIS

INR AGAINST USD (1999-2023)

TIME SERIES ANALYSIS PROJECT



OUR TEAM



Jenithrika



Mamta



Paavni



Rashi



Rishika

Time Series Analysis

ACKNOWLEDGEMENT

We would like to thank Dr. Renu Kaul for giving us the opportunity, help, and guidance to complete this analysis project on exchange rate. It was an excellent learning experience. We would like to take this opportunity to thank all those who worked hard to teach us and those who helped us to successfully complete this project. Additionally, the project would not have been successful without the cooperation and contributions of all the team members.

Thank You



WHY THIS TOPIC ?

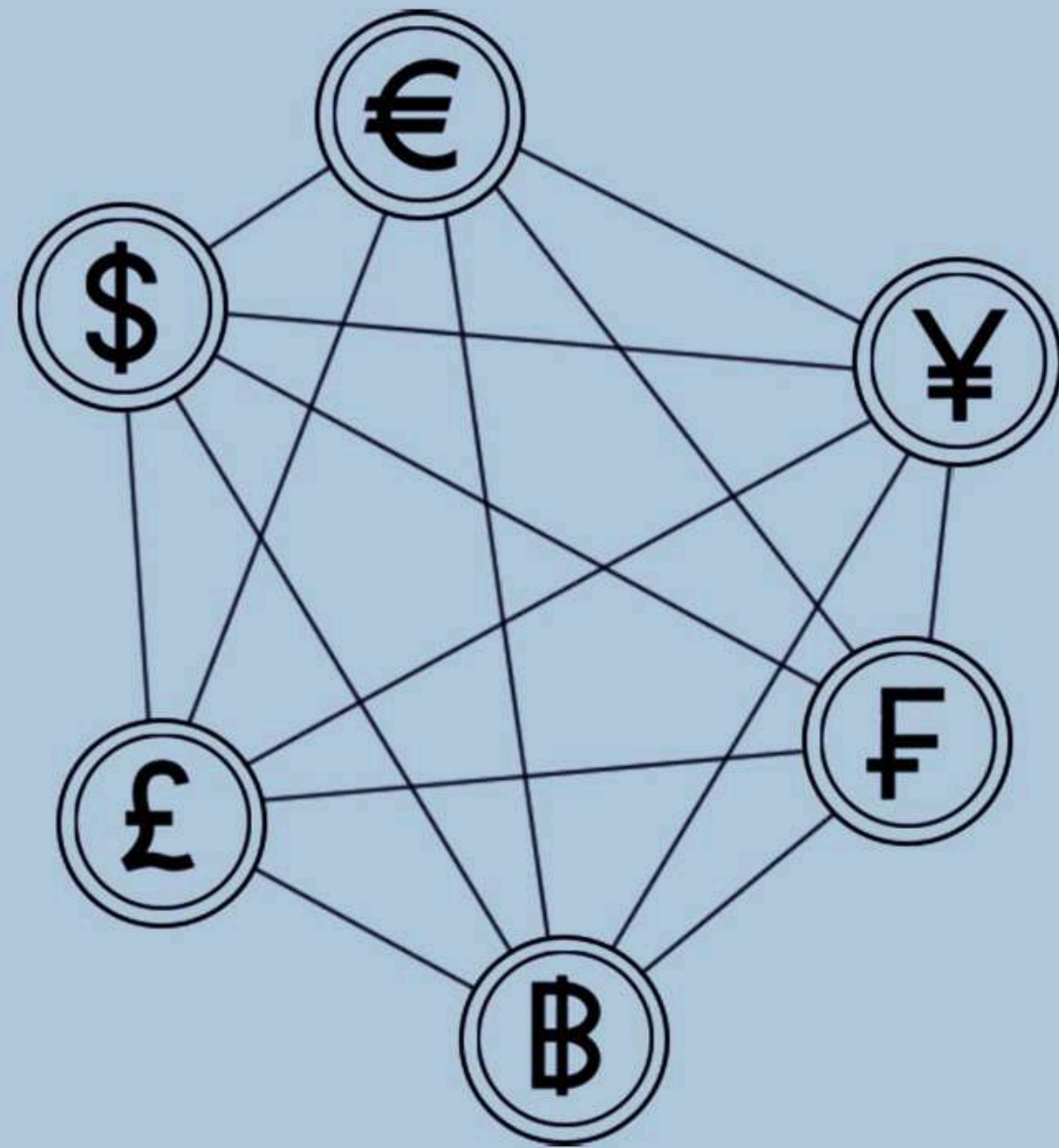
Exchange rates are a reflection of a country's economic health and global economic events. Having a basic understanding of exchange rates can help individuals stay informed about the economic context in which they live and make sense of financial news.

Understanding exchange rates can help individuals make more informed financial decisions, such as when making investments, taking out loans, or managing their personal finances.

Even for common individuals who are not studying finance or economics academically, a basic understanding of exchange rates can be beneficial for everyday financial decisions, international transactions, and staying economically informed in a globally connected world. It can help people make better financial choices and manage their resources more effectively.



BASIC CONCEPTS



Exchange Rate

['iks-'chānj 'rāt]

An exchange rate is a rate at which one currency will be exchanged for another currency and affects trade and the movement of money between countries.

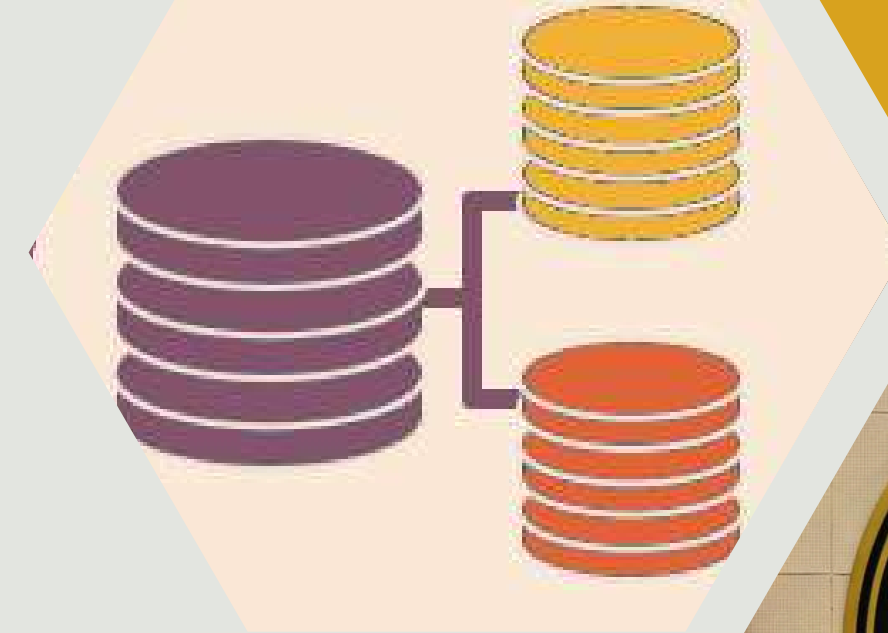


TYPES OF EXCHANGE RATE

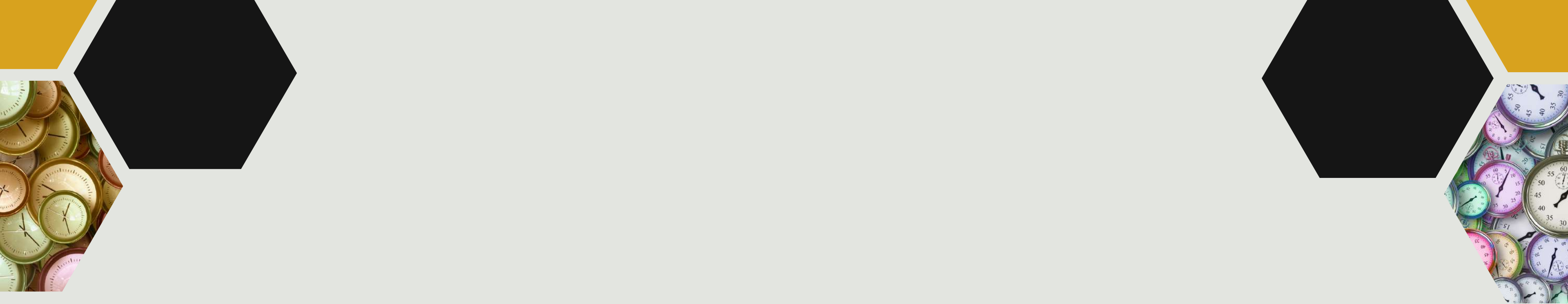
1. **Free floating rate** - determined by market forces of supply and demand, USA, China, Japan
2. **Fixed exchange rate** - controlled by the central bank of an economy, usually pegged against a currency, UAE, Hong Kong
3. **Managed Floating** - Mix of both the above systems. (Adopted by India)



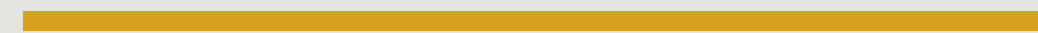
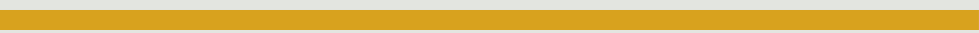
ABOUT DATASET



- Our dataset was sourced from the **Reserve Bank of India**.
- We collected data from January 1, 1999, up to October 30, 2023.
- The dataset primarily comprises **daily exchange rate values**, representing the cost of 1 US dollar in Indian rupees.
- The data is collected for working days, which gives us approximately **6,000+** data points to work with.



ANALYSIS



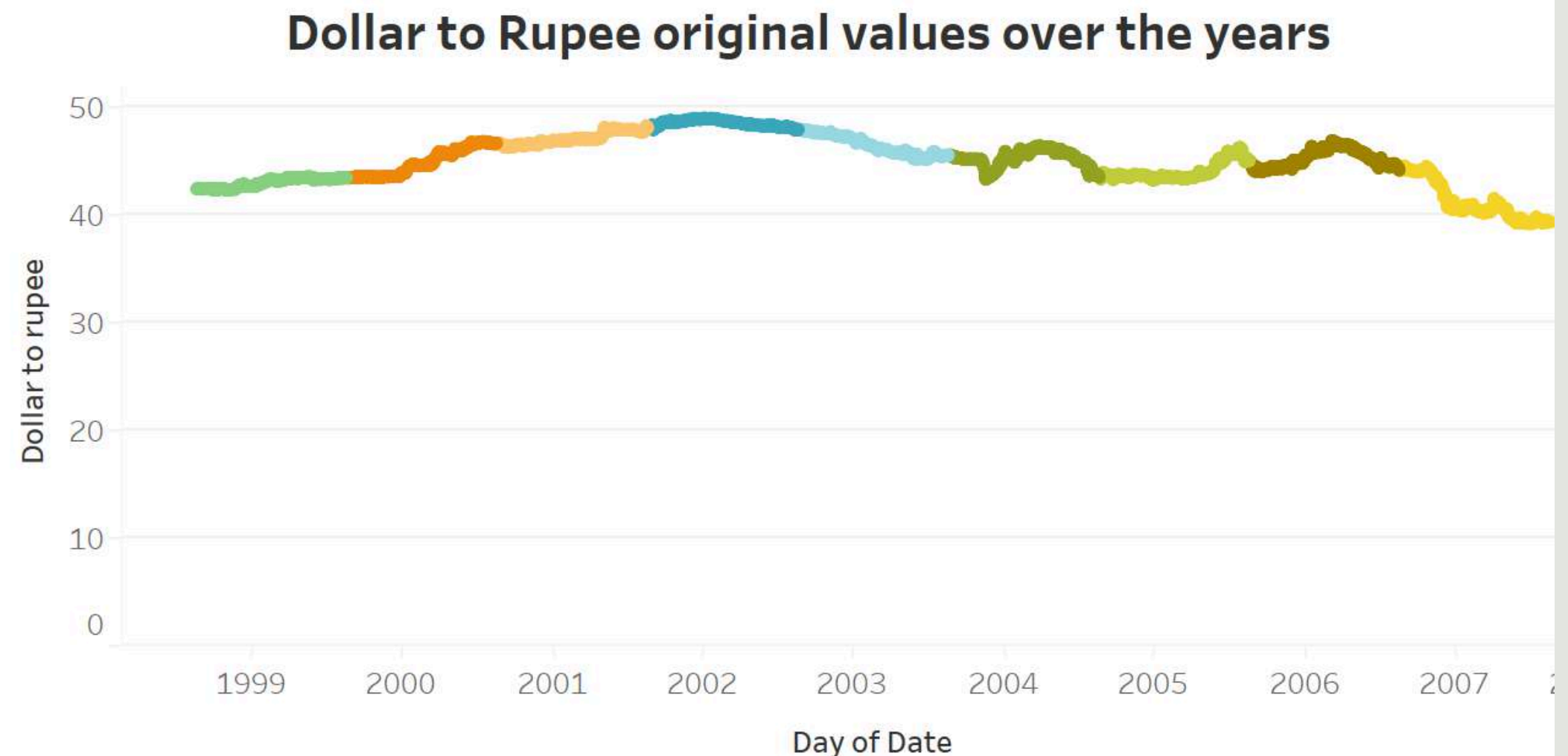
SITUATION IN INDIA (1999-2007)

1. 1999-2002: Stable Exchange Rate:

- **No major fluctuations** witnessed
- Minor fluctuations mainly because of **regular** global events

2. 2002-2007: Appreciation of the Rupee:

- Partly due to **rising economic growth** in India
- **Booming IT industry**, foreign **investment inflows**
- Rate moved from 48INR to 40 INR



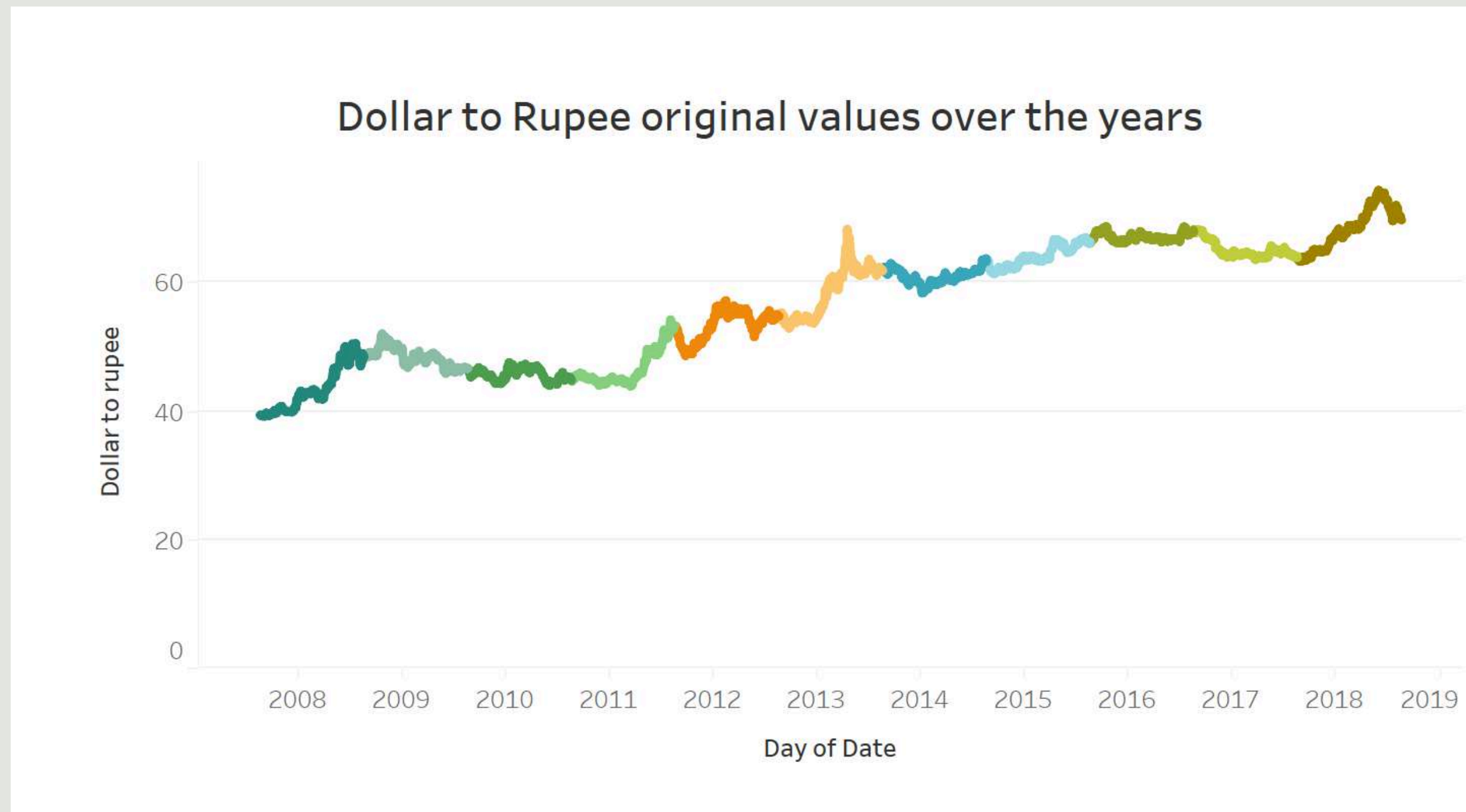
SITUATION IN INDIA (2008-2018)

3. 2008-2013: Periods of Volatility:

- **Global Financial Crisis of 2008:** investors became risk averse, capital flight
- **RBI** stepped in to control the situation

4. 2013-2018: Depreciation of the Rupee:

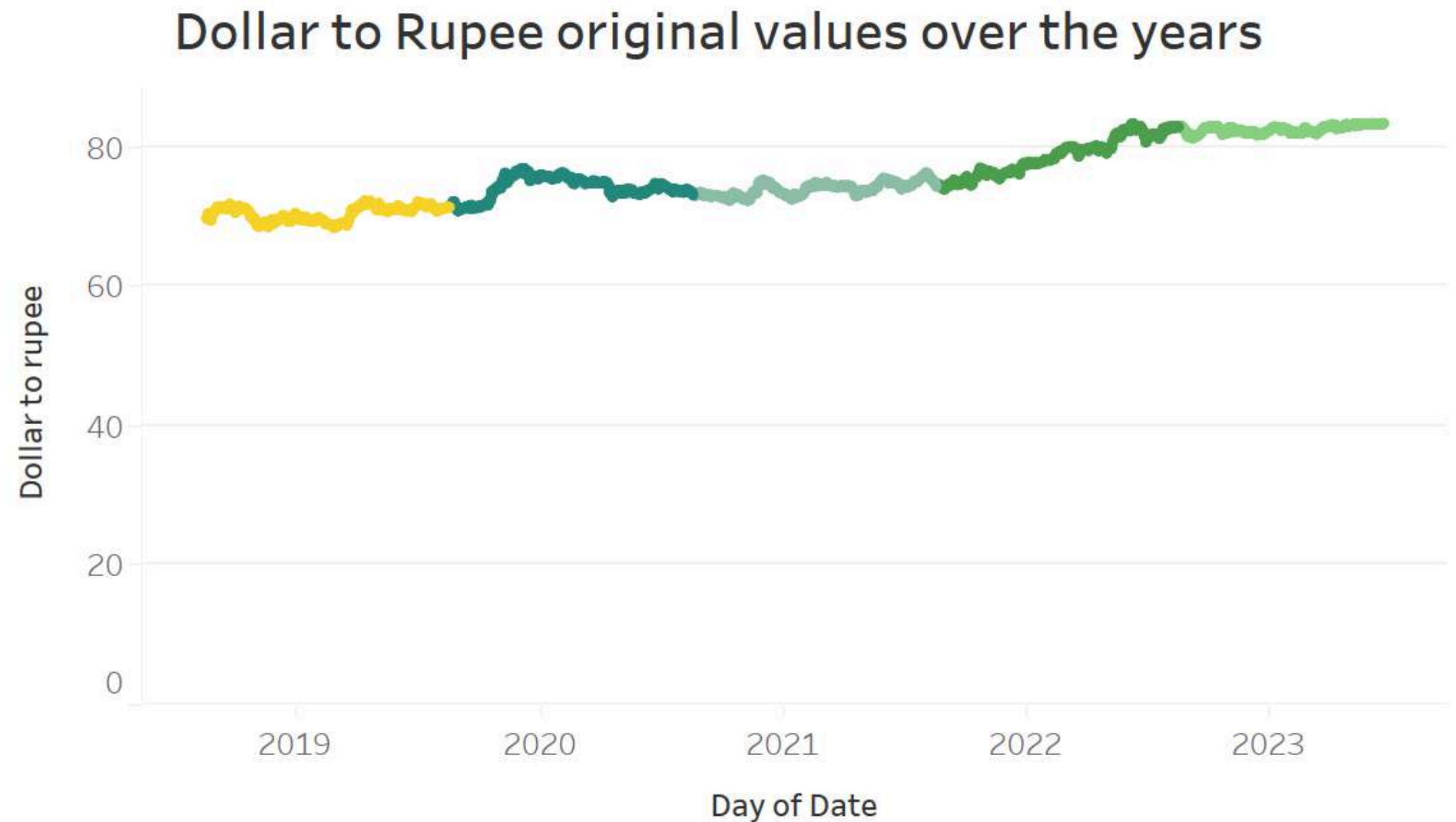
- Growing **Current account deficit**
- Rate moved from **55 INR to 68 INR.**



SITUATION IN INDIA (2019-PRESENT)

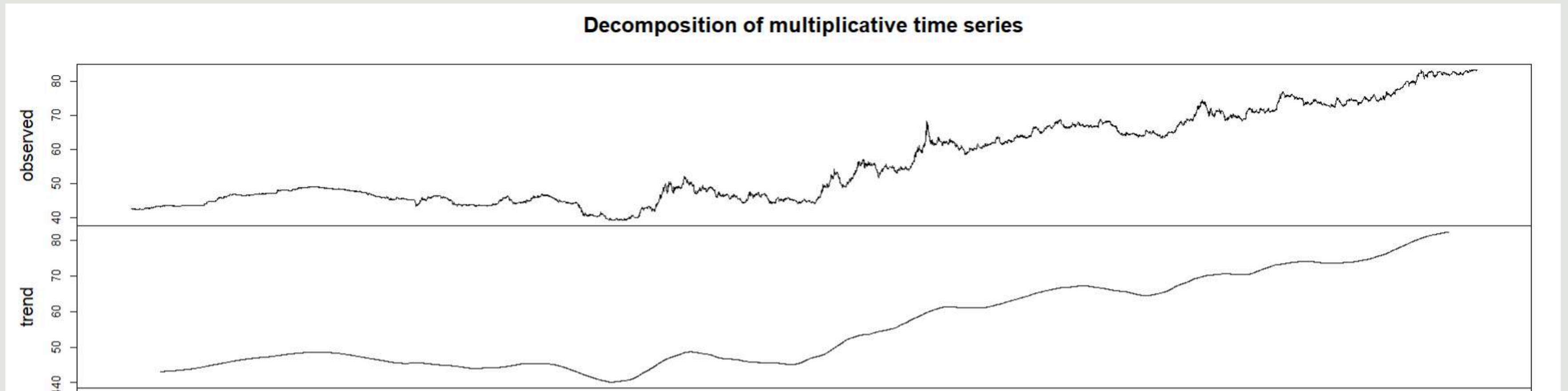
5. 2018-2022: Mixed Trends:

- **Mixed trends** though the rate has gone up from 68 INR to 83 INR in these years
- **Trade tensions, interest rate differentials, COVID-19**
- **RBI managed volatility**



DECOMPOSITION OF TIME SERIES

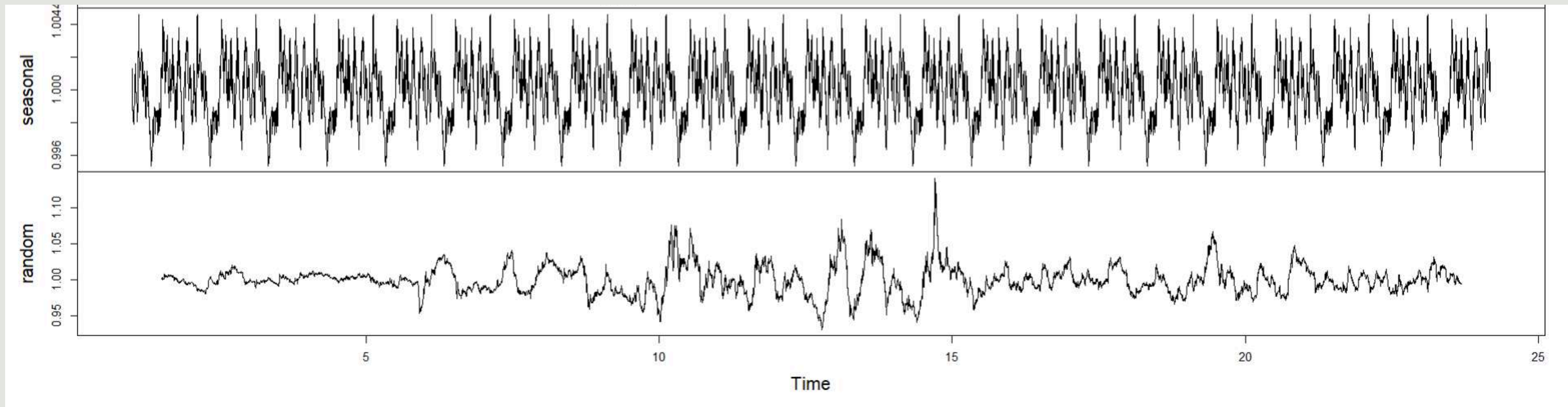
Trend



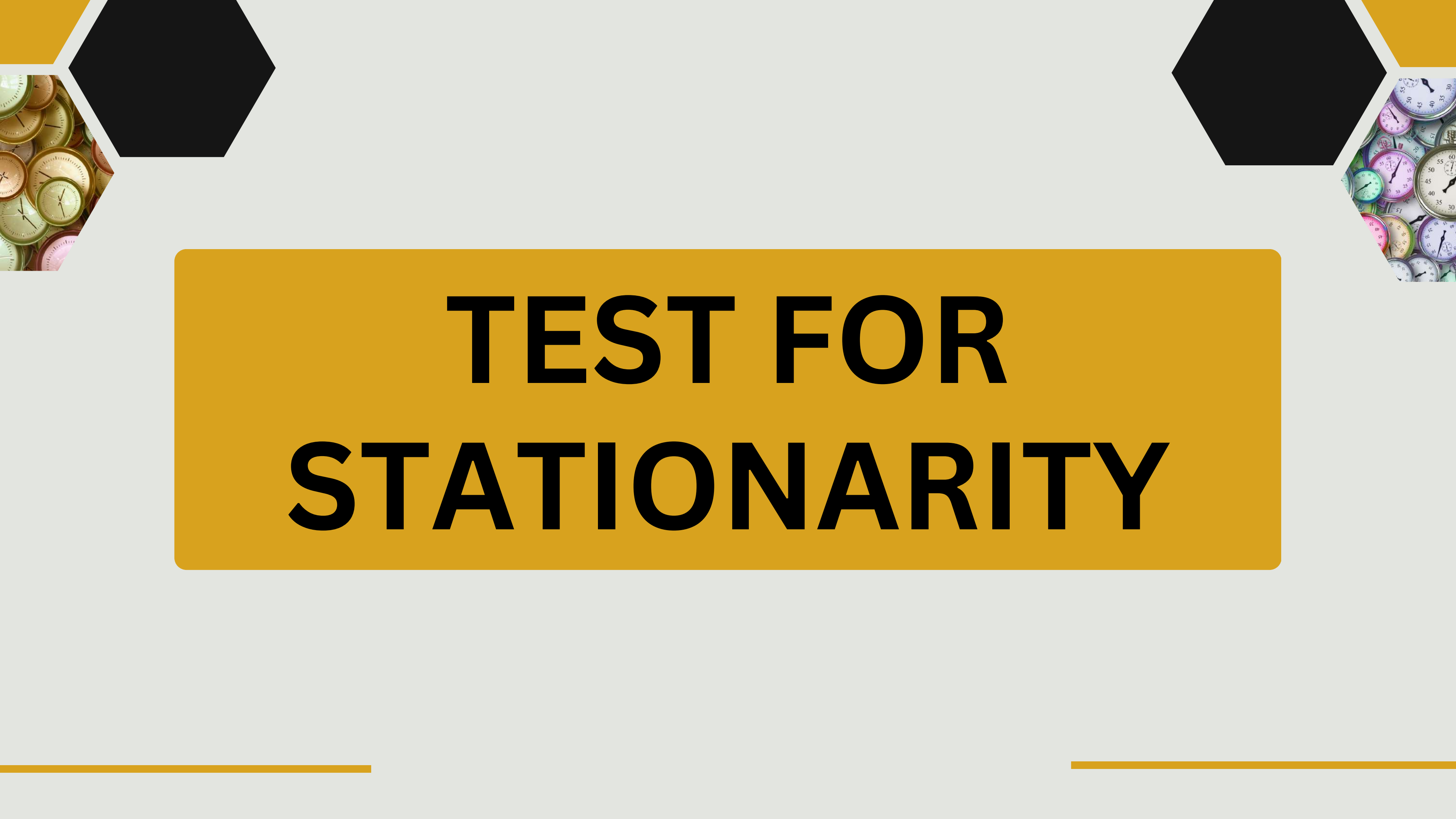
Economic factors, Tech advancements, Govt policies can cause the trend variations being observed in the data

DECOMPOSITION OF TIME SERIES

Seasonal Variations and Random Component



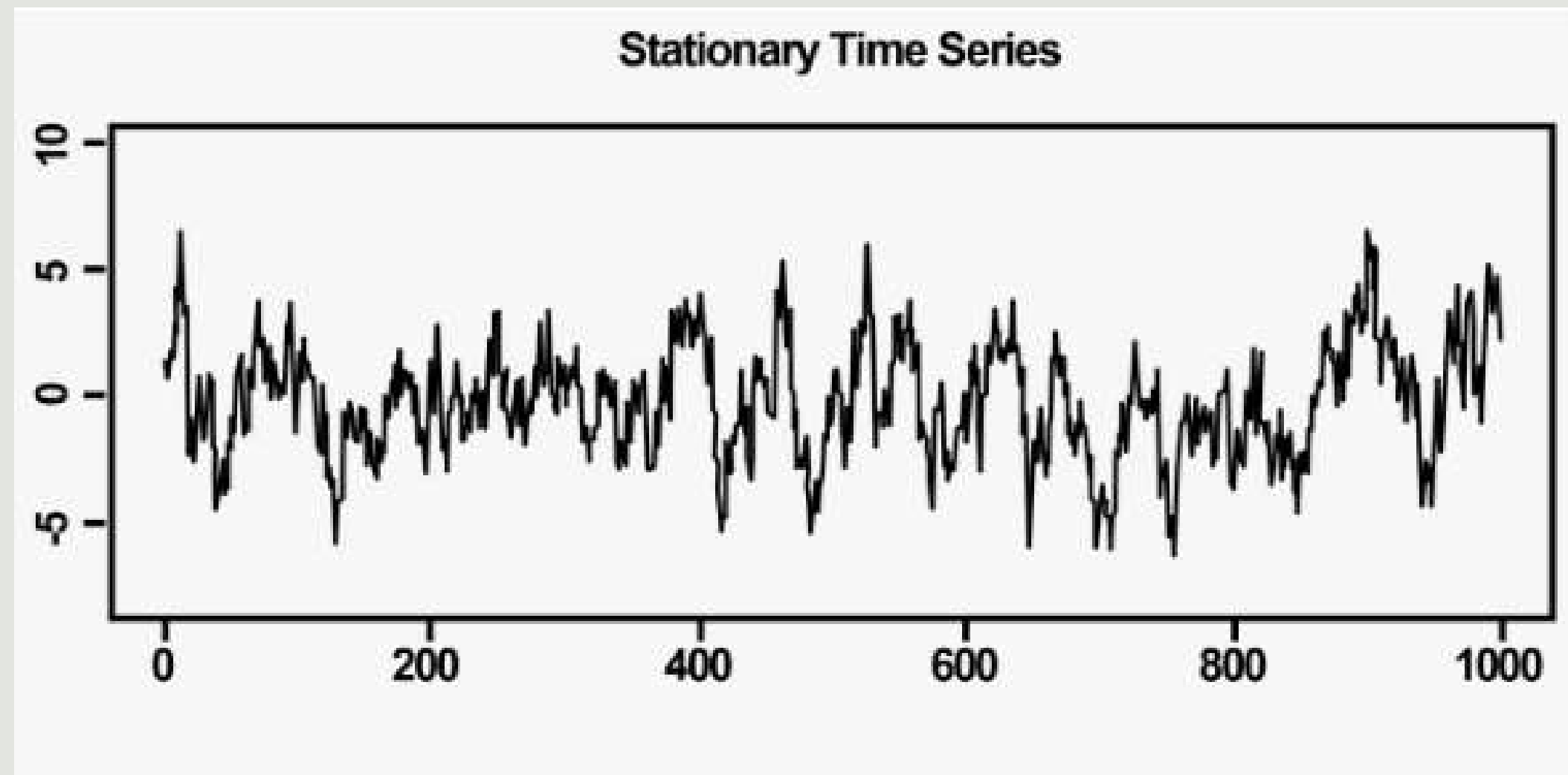
Interest rate policies, Geopolitical events, Trade seasons may influence seasonal variations while unanticipated events give rise to random component.



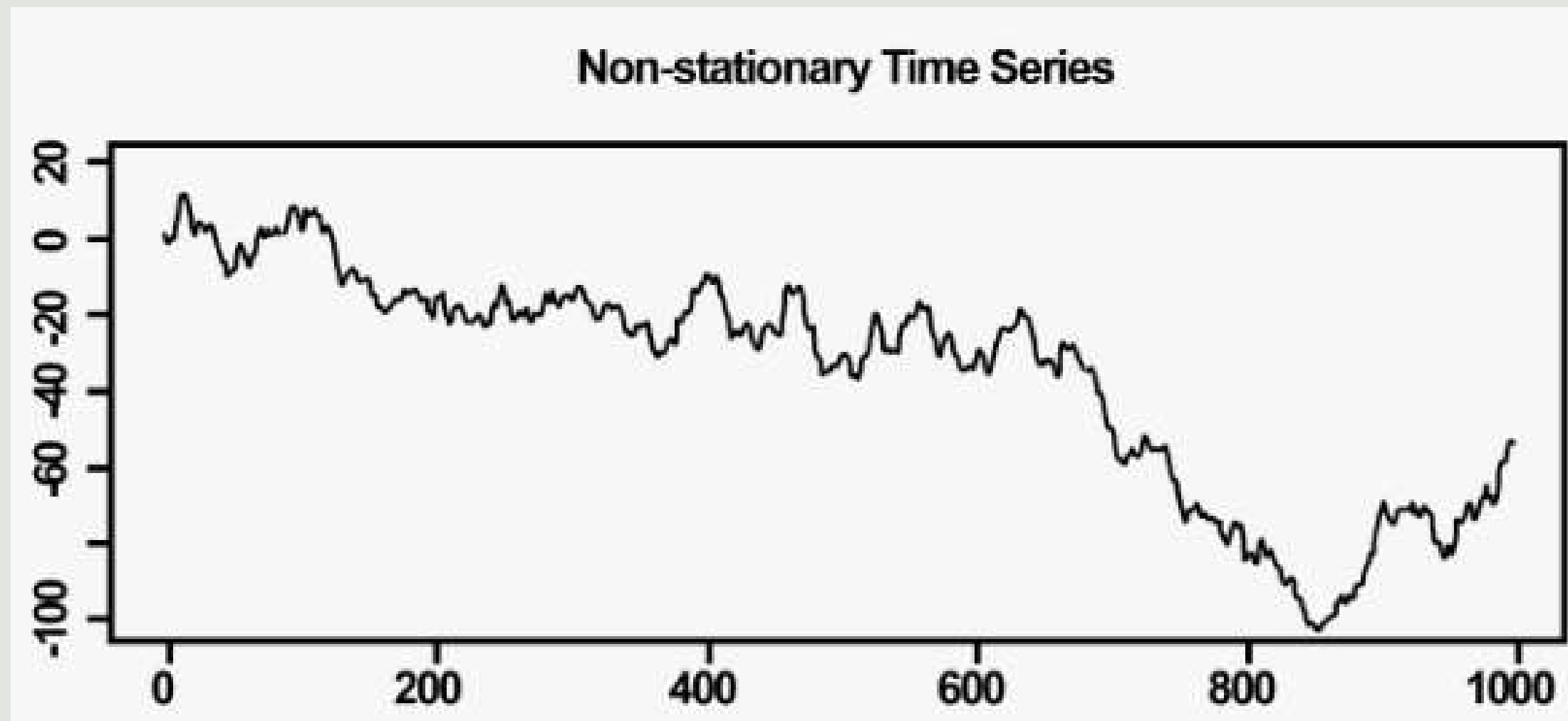
TEST FOR STATIONARITY

STATIONARITY

- A time series is said to be **stationary** if its statistical properties, such as mean, variance doesn't change over time.



- A **non-stationary** time series is a time series data set where the statistical properties of the data change over time. Non-stationary time series typically exhibit trends, seasonality, or other systematic patterns that evolve over time.



WHY STATIONARY TIME SERIES ?

- **Simplified Analysis:** In a stationary series, statistical properties like mean and variance remain constant over time. This simplifies modeling and forecasting.
- **Statistical Assumptions:** Many time series models and techniques, such as ARIMA models assume stationarity. When the data is non-stationary, it may violate these assumptions and lead to inaccurate results.
- **Predictability:** Stationary time series are often more predictable because they exhibit consistent statistical properties. Non-stationary series can exhibit trends and cycles that make forecasting more challenging.

HOW TO MAKE A TIME SERIES STATIONARY ?



ADF TEST

The ADF test is one of the most commonly used tests for checking the Stationarity. It is based on the idea of testing whether the coefficient of the lagged level of the time series is equal to one in a regression equation. It helps determine whether differencing is necessary to make the data stationary.

consider the following model equation for auto regressive(p) model:

$$y_t = c + \beta t + \alpha y_{t-1} + \phi_1 \Delta Y_{t-1} + \phi_2 \Delta Y_{t-2} \dots + \phi_p \Delta Y_{t-p} + e_t$$

Here, $Y(t-1)$ = lag 1 a time series and $\phi(\Delta) Y(t-p)$, $p \geq 1$ is p difference of time series at time(t-p).

An ADF test is a unit root test that tests the null hypothesis that $\alpha = 1$ in the following model equation. α (alpha) is the coefficient of the first lag on Y. If not rejected, the series is taken to be non-stationary.



ADF TEST

The hypothesis for the test:

The null hypothesis, **H0**: There is a unit root. i.e., The time series is non-stationary.

The alternative hypothesis, **H1**: The time series is stationary.

If the p-value is less than 5% then we reject the null hypothesis and conclude that unit root is not there. If the p-value is more than 5% then we accept the null hypothesis hence the data is non stationary.

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.2236 on 6026 degrees of freedom
```

```
Multiple R-squared:  0.0007036, Adjusted R-squared:  0.0003719
```

```
F-statistic: 2.121 on 2 and 6026 DF,  p-value: 0.12
```



KPSS TEST

The KPSS test is a statistical test used in time series analysis to check for the stationarity of a time series data set.

It checks for the presence of departures from stationarity due to deterministic trends or around a constant mean. It helps assess whether the data is stationary around a trend.

The hypothesis for the test

Null Hypothesis (**H0**): Series is trend stationary or series has no unit root.

Alternate Hypothesis (**H1**): Series is non-stationary, or series has a unit root.

Note: The hypothesis is reversed in the KPSS test compared to ADF Test.

If the null hypothesis is failed to be rejected, this test may provide evidence that the series is trend stationary



KPSS TEST

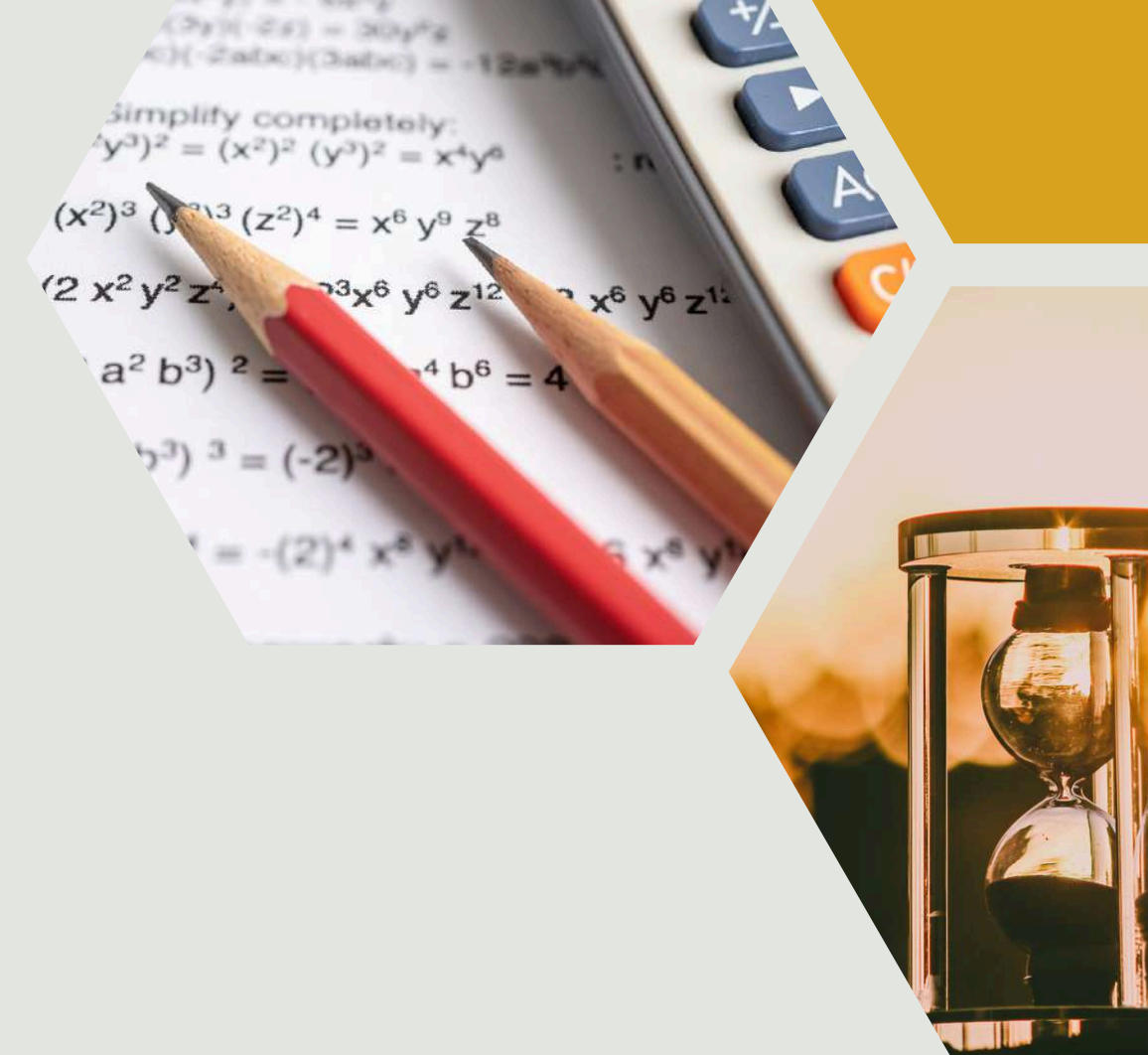
```
#####  
# KPSS Unit Root Test #  
#####
```

Test is of type: tau with 11 lags.

Value of test-statistic is: 9.2964

Critical value for a significance level of:

	10pct	5pct	2.5pct	1pct
critical values	0.119	0.146	0.176	0.216

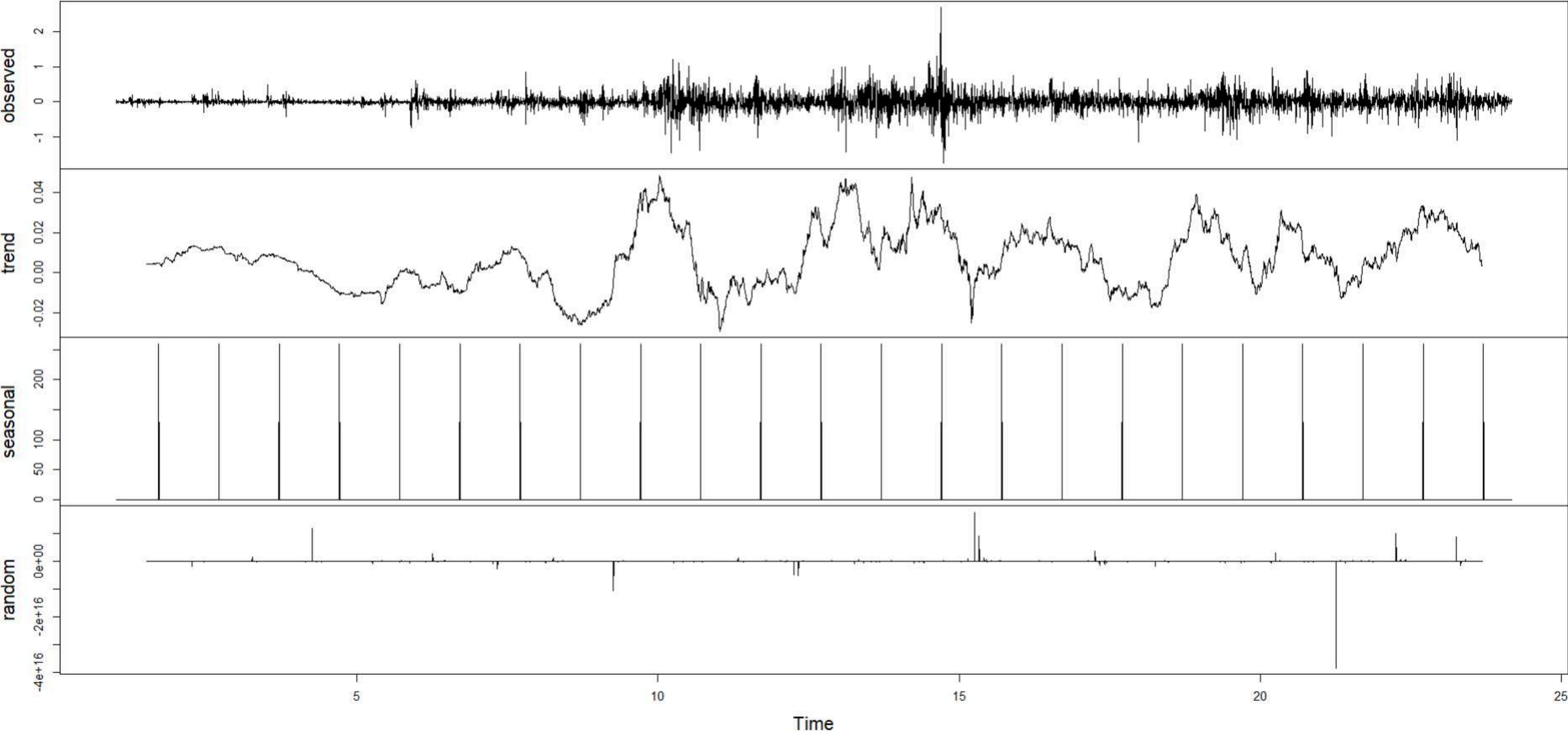


WHAT NEXT?



Need to difference the data to make it
Stationary!

Decomposition of multiplicative time series



ADF TEST

The hypothesis for the test:

The null hypothesis, **H0**: There is a unit root. i.e., The time series is non-stationary.

The alternative hypothesis, **H1**: The time series is stationary.

If the p-value is less than 5% then we reject the null hypothesis and conclude that unit root is not there. If the p-value is more than 5% then we accept the null hypothesis hence the data is non stationary.

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.2236 on 6023 degrees of freedom
```

```
Multiple R-squared:  0.5055,    Adjusted R-squared:  0.5053
```

```
F-statistic:  2053 on 3 and 6023 DF,  p-value: < 2.2e-16
```



KPSS TEST

The hypothesis for the test

Null Hypothesis (**H0**): Series is trend stationary or series has no unit root.

Alternate Hypothesis (**H1**): Series is non-stationary, or series has a unit root.

```
#####  
# KPSS Unit Root Test #  
#####
```

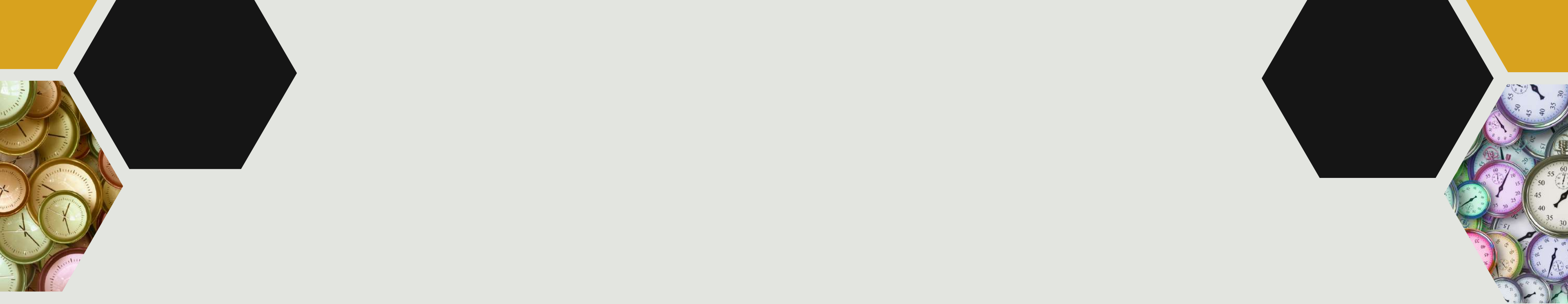
```
Test is of type: tau with 11 lags.
```

```
Value of test-statistic is: 0.0388
```

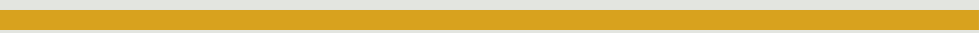
```
Critical value for a significance level of:
```

	10pct	5pct	2.5pct	1pct
critical values	0.119	0.146	0.176	0.216





MODEL FITTING



WHY ARIMA?

The ARIMA model (Autoregressive integrated moving average model) is the generalization of the autoregressive moving average (ARMA) model.

ARIMA models are applied in cases where data shows evidence of non stationarity:

- In the sense of Mean (but not variance/autocovariance)
- Where an initial differencing step can be applied one or more times to eliminate the non-stationarity of the mean function (i.e. the trend)



WHY ARIMA?

AR part of ARIMA indicates that the evolving variable of interest is regressed on its own lagged (i.e. prior) values whereas the moving average part (MA) part indicates that regression error is actually a linear combination of error terms whose values occurred contemporaneously and at various times in the past.

ARIMA models are denoted by $ARIMA(p,d,q)$ where

- p is the order or number of time lags or order of auto regressive (AR) term (number of past values one future value depends on)
- d is the degree of differencing in order to make the data stationary
- q is the order (extent) of the moving average model



WHY ARIMA?

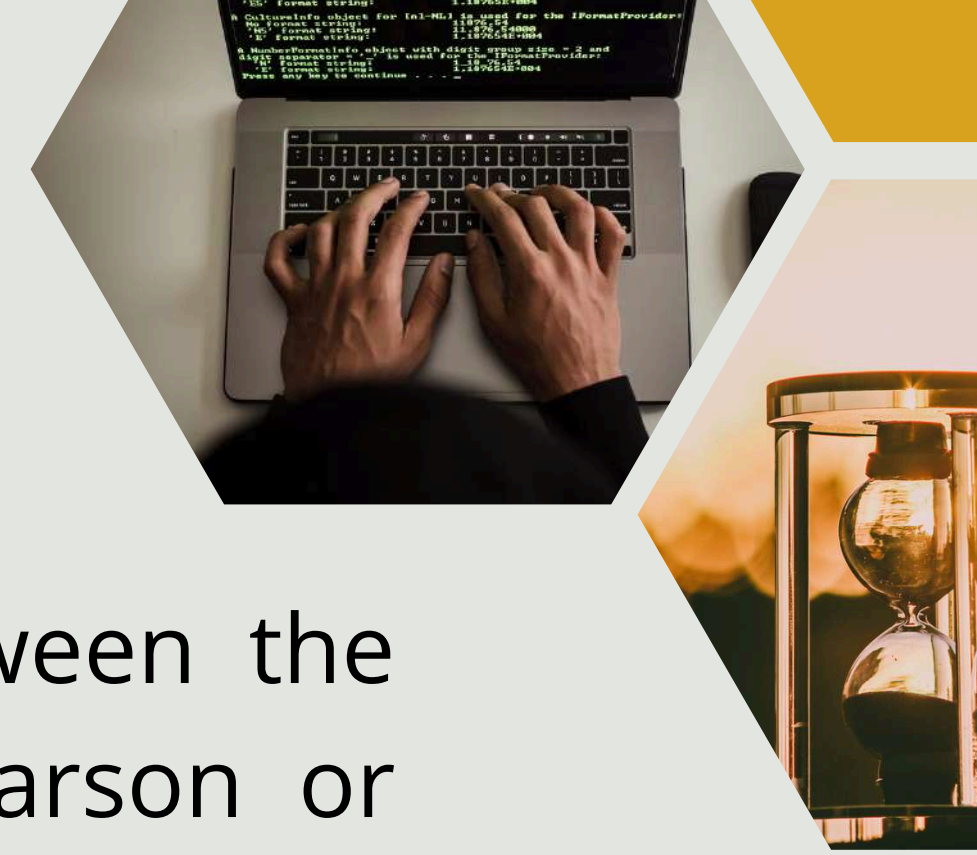
- ARIMA models address non-stationarity, transforming data to achieve stationarity, essential for modeling.
- The Autoregressive (AR) component of ARIMA models captures past dependencies in exchange rate movements.
- The Moving Average (MA) component of ARIMA models considers past errors or shocks, accounting for their influence on current rates.



ACF AND PACF

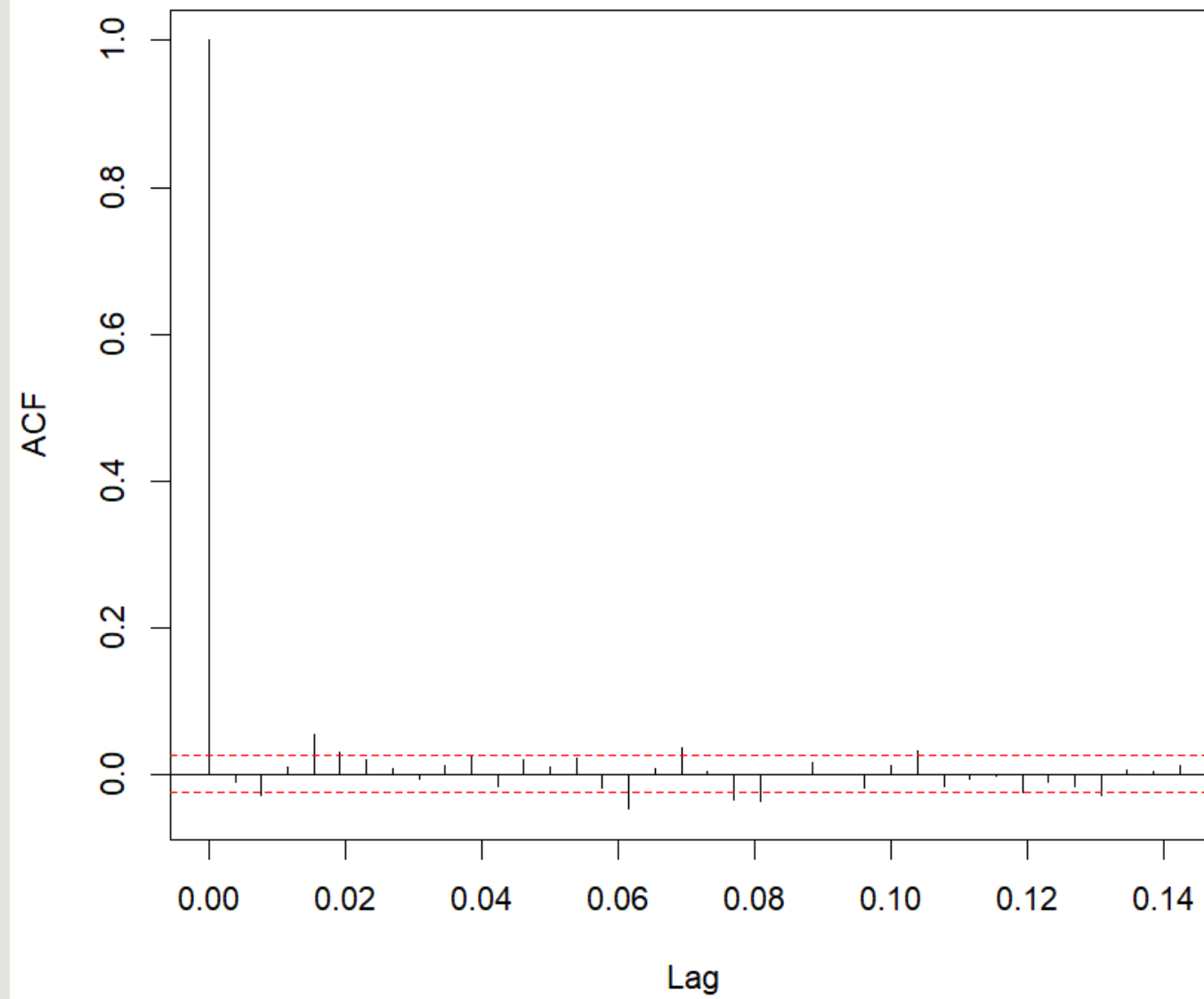
ACF plot is the graph showing the autocorrelation between the variable and its lag value. It is either calculated by Pearson or spearman rank correlation function. ACF plot is used to find out the order of MA process. The ACF starts at a lag of 0, which is the correlation of the time series with itself and therefore results in a correlation of 1.

PACF plot is the graph showing the partial autocorrelation between the variable and its lag value controlling the effect of other lag values. PACF plot is used to find out the order of AR Process.

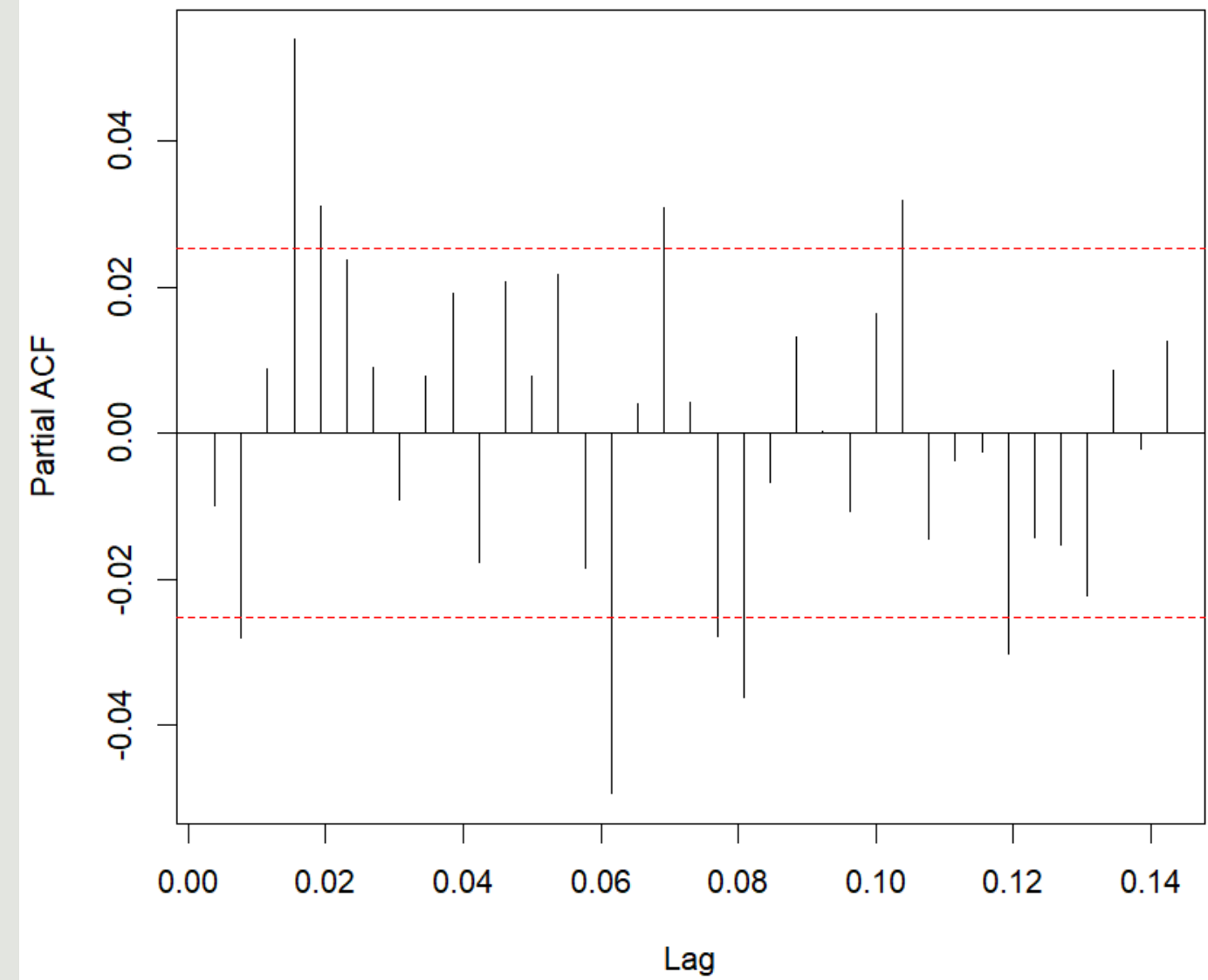


ACF AND PACF GRAPH

ACF with 95% Confidence Intervals



PACF with 95% Confidence Intervals



AKAIKE'S INFORMATION CRITERIA (AIC)

AIC - Akaike's Information Criterion (AIC) is the most commonly used and is given (approximately) by:

AIC = $-2 \ln(\text{max. likelihood}) + 2r$ where r denotes the number of independent parameters that are fitted for the model being assessed. Thus the AIC essentially chooses the model with the best fit, as measured by the likelihood function, subject to a penalty term, to prevent over-fitting, that increases with the number of parameters in the model.



AKAIKE'S INFORMATION CRITERIA (AIC)

When fitting models, it is possible to increase the maximum likelihood by adding parameters, but doing so may result in overfitting. Both AIC (Akaike information criterion) and BIC (Bayesian information criterion) attempt to resolve this problem by introducing a penalty term for the number of parameters in the model, although BIC generally penalizes free parameters more strongly than the AIC.

$$\text{BIC} = -2 \ln(\text{max. likelihood}) + r \ln(n)$$



AIC VALUES FOR DIFFERENT MODELS

AIC: -942.2457 Model: ARIMA(0 , 1, 0)
AIC: -940.7671 Model: ARIMA(0 , 1, 1)
AIC: -942.6666 Model: ARIMA(0 , 1, 2)
AIC: -941.6278 Model: ARIMA(0 , 1, 3)
AIC: -959.419 Model: ARIMA(0 , 1, 4)
AIC: -940.739 Model: ARIMA(1 , 1, 0)
AIC: -939.3282 Model: ARIMA(1 , 1, 1)
AIC: -940.8406 Model: ARIMA(1 , 1, 2)
AIC: -959.1668 Model: ARIMA(1 , 1, 3)
AIC: -965.1956 Model: ARIMA(1 , 1, 4)
AIC: -943.1694 Model: ARIMA(2 , 1, 0)
AIC: -941.2886 Model: ARIMA(2 , 1, 1)
AIC: -963.189 Model: ARIMA(2 , 1, 2)
AIC: -961.6509 Model: ARIMA(2 , 1, 3)

AIC: -963.376 Model: ARIMA(2 , 1, 4)
AIC: -941.7484 Model: ARIMA(3 , 1, 0)
AIC: -958.1496 Model: ARIMA(3 , 1, 1)
AIC: -961.6745 Model: ARIMA(3 , 1, 2)
AIC: -956.9058 Model: ARIMA(3 , 1, 3)
AIC: -961.5039 Model: ARIMA(3 , 1, 4)
AIC: -957.8257 Model: ARIMA(4 , 1, 0)
AIC: -965.7771 Model: ARIMA(4 , 1, 1)
AIC: -963.7781 Model: ARIMA(4 , 1, 2)
AIC: -961.5907 Model: ARIMA(4 , 1, 3)
AIC: -959.9831 Model: ARIMA(4 , 1, 4)
AIC: -961.9405 Model: ARIMA(5 , 1, 0)
AIC: -963.7797 Model: ARIMA(5 , 1, 1)
AIC: -961.7932 Model: ARIMA(5 , 1, 2)
AIC: -959.8066 Model: ARIMA(5 , 1, 3)
AIC: -957.8778 Model: ARIMA(5 , 1, 4)

BIC VALUES FOR DIFFERENT MODELS

BIC: -935.5413 Model: ARIMA(0 , 1, 0)
BIC: -927.3584 Model: ARIMA(0 , 1, 1)
BIC: -922.5536 Model: ARIMA(0 , 1, 2)
BIC: -914.8105 Model: ARIMA(0 , 1, 3)
BIC: -925.8973 Model: ARIMA(0 , 1, 4)
BIC: -927.3303 Model: ARIMA(1 , 1, 0)
BIC: -919.2152 Model: ARIMA(1 , 1, 1)
BIC: -914.0233 Model: ARIMA(1 , 1, 2)
BIC: -925.6451 Model: ARIMA(1 , 1, 3)
BIC: -924.9696 Model: ARIMA(1 , 1, 4)
BIC: -923.0564 Model: ARIMA(2 , 1, 0)
BIC: -914.4713 Model: ARIMA(2 , 1, 1)
BIC: -929.6673 Model: ARIMA(2 , 1, 2)
BIC: -921.4249 Model: ARIMA(2 , 1, 3)
BIC: -916.4456 Model: ARIMA(2 , 1, 4)

BIC: -914.931 Model: ARIMA(3 , 1, 0)
BIC: -924.628 Model: ARIMA(3 , 1, 1)
BIC: -921.4484 Model: ARIMA(3 , 1, 2)
BIC: -909.9755 Model: ARIMA(3 , 1, 3)
BIC: -907.8692 Model: ARIMA(3 , 1, 4)
BIC: -924.304 Model: ARIMA(4 , 1, 0)
BIC: -925.5511 Model: ARIMA(4 , 1, 1)
BIC: -916.8477 Model: ARIMA(4 , 1, 2)
BIC: -907.956 Model: ARIMA(4 , 1, 3)
BIC: -899.644 Model: ARIMA(4 , 1, 4)
BIC: -921.7145 Model: ARIMA(5 , 1, 0)
BIC: -916.8493 Model: ARIMA(5 , 1, 1)
BIC: -908.1585 Model: ARIMA(5 , 1, 2)
BIC: -899.4675 Model: ARIMA(5 , 1, 3)
BIC: -890.8344 Model: ARIMA(5 , 1, 4)

OPTIMAL ARIMA MODEL

```
[[1]]$AIC
```

```
[1] -965.7771
```

```
[[1]]$BIC
```

```
[1] -925.5511
```

```
[[1]]$Description
```

```
[1] "ARIMA( 4 , 1, 1 )"
```

OUR ARIMA MODEL

```
arima(x = ts(A, start(1), frequency = 1), order = c(4, 1, 1))
```

Coefficients:

	ar1	ar2	ar3	ar4	ma1
	0.5786	-0.0202	0.0262	0.0492	-0.5910
s.e.	0.1279	0.0149	0.0152	0.0149	0.1279

sigma^2 estimated as 0.04978: log likelihood = 488.89, aic = -965.78

Training set error measures:

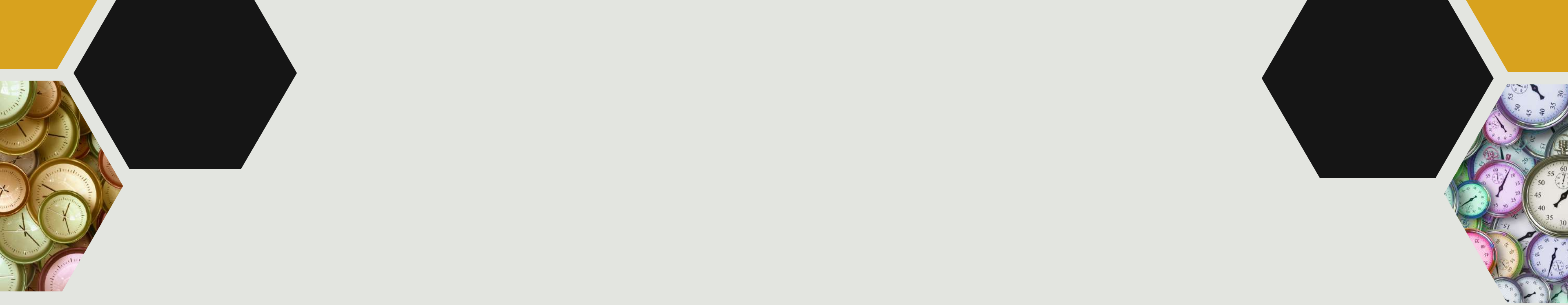
	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	0.006060459	0.2231057	0.1430419	0.009324573	0.2525911	0.9992346

ACF1

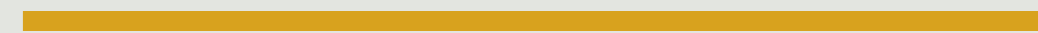
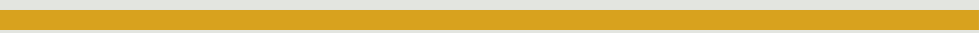
Training set	-0.0007181443
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$$X_t = 0.5786X_{t-1} - 0.0202X_{t-2} + 0.0262X_{t-3} + 0.0492X_{t-4} + Z_t - (-0.5910) Z_{t-1}$$





FORECASTING

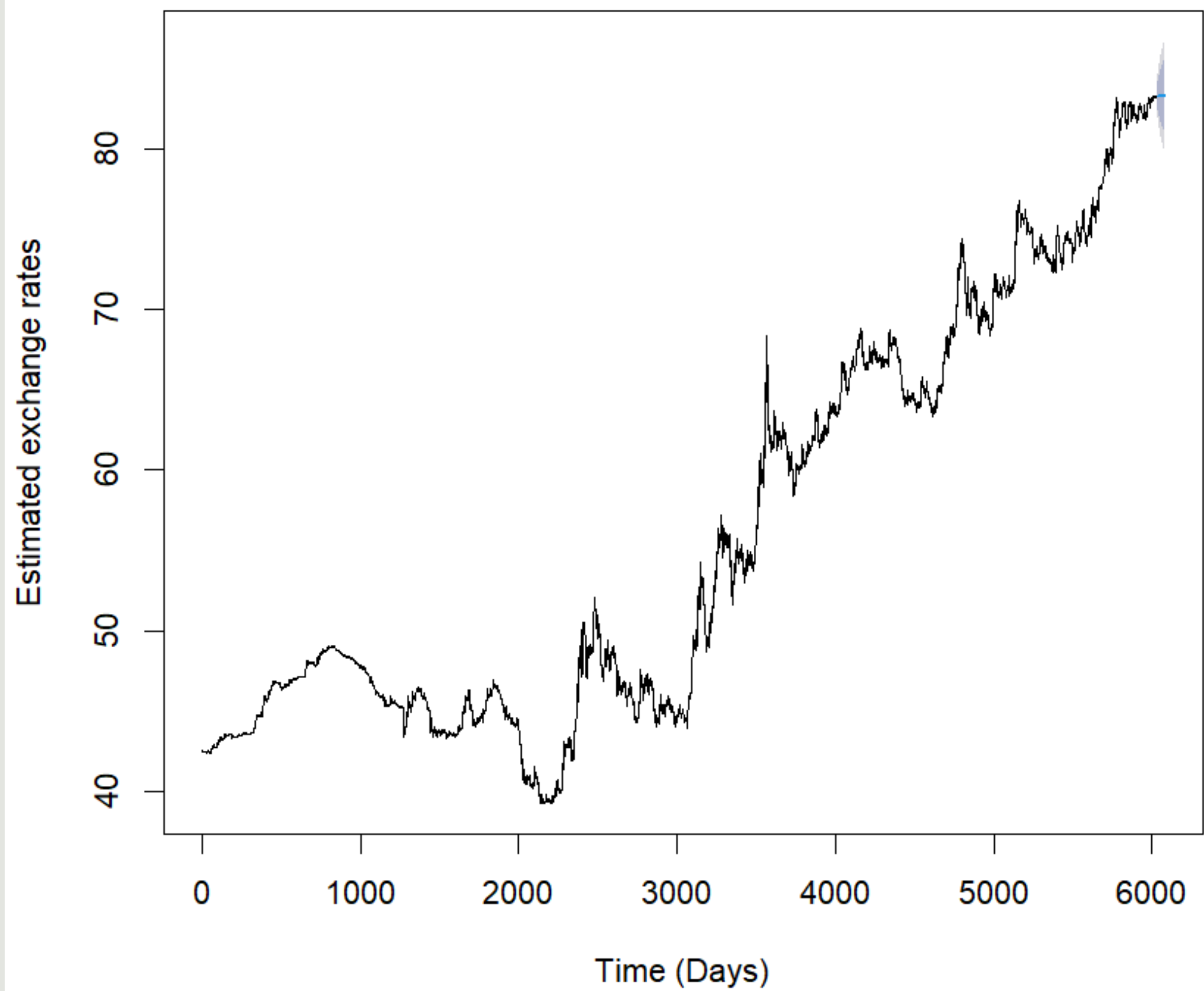


FORECASTING

- Our **data source** was RBI Reference Exchange Rate of Indian Rupee against US Dollar. We collected daily exchange rates with from 1st January 1990 to 31st October 2023 with total data points coming up to be 6055 points.
- The ARIMA model is (4,1,1)



Forecasts from ARIMA(4,1,1)



FORECASTED VALUES

```
[1] 83.27261 83.27514 83.27753 83.27893 83.27989 83.28060 83.28115 83.28154 83.28183
[10] 83.28203 83.28218 83.28229 83.28237 83.28243 83.28247 83.28251 83.28253 83.28254
[19] 83.28256 83.28257 83.28257 83.28258 83.28258 83.28258 83.28258 83.28258 83.28259
```

**Did you notice anything out
of the ordinary?**

FORECASTED VALUES



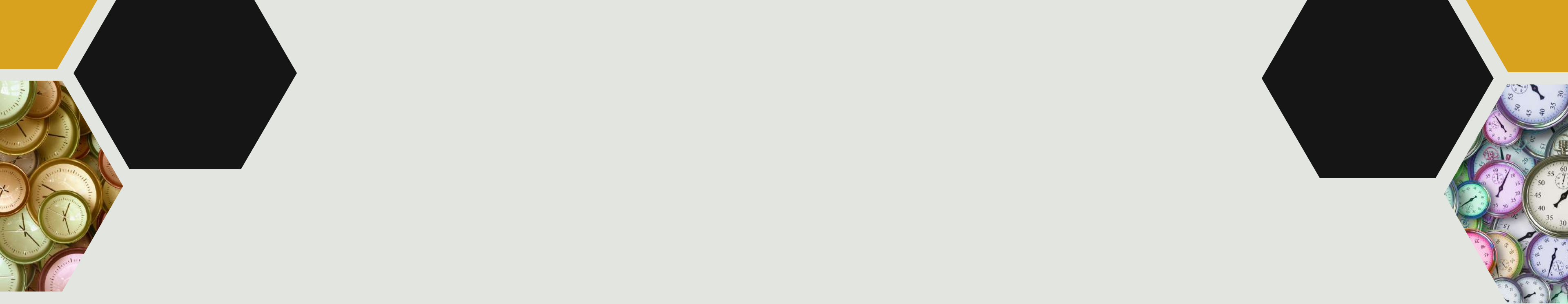
Now were we able to predict the future?

DATE	PREDICTED VALUE	ACTUAL VALUE
NOVEMBER 1, 2023	83.2761 ~ 83.28	83.28
NOVEMBER 2, 2023	83.27514 ~ 83.26	83.26
NOVEMBER 3, 2023	83.27753 ~ 83.28	83.27
NOVEMBER 6, 2023	83.27893 ~ 83.28	83.22

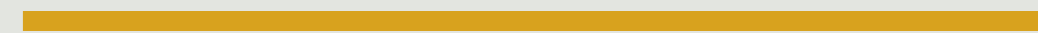
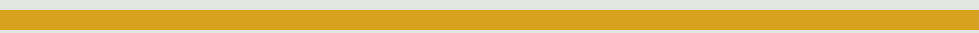
RESIDUALS

```
[1]  4.250998e-02  4.986163e-02 -4.940895e-02  1.086350e-02 -1.748141e-03
[6] -1.979503e-03 -1.896389e-02 -1.237103e-04  1.952191e-02 -4.951190e-02
[11]  3.106028e-02 -5.374638e-04  1.061311e-02  2.161879e-03  2.910640e-06
[16] -2.026004e-02  9.107141e-03 -8.085127e-04  2.478869e-04  8.692853e-04
[21] -1.997857e-02 -2.023424e-02  9.210345e-03 -2.239981e-04 -2.842213e-02
[26]  1.284891e-03  1.966056e-02 -9.168402e-03 -2.775086e-02  7.023311e-02
[31] -3.273453e-04 -3.750076e-02 -9.372362e-03 -4.400714e-02  3.798302e-02
[36]  5.072409e-02  3.393146e-03  3.938082e-03  1.904932e-02  3.722370e-02
```

Too small residuals which implies better fit!



DIAGNOSTIC CHECK



DIAGNOSTIC CHECK

- Diagnostic checking is a critical step in evaluating the performance.
- The primary goal of diagnostic checking is to identify any issues or shortcomings in the chosen model and to make necessary adjustments to improve its accuracy and predictive power.
- Here we examined the model's residuals from the fitted model to ensure its reliability.
- If we have a 'good' model, then we expect the residuals to be 'random' and 'close to zero', and model validation tools are used to see whether this is the case.



ACCURACY MEASURES

To assess forecast accuracy, we used multiple metrics:

- **Mean Error (ME):** The average difference between predicted and actual values.

$$ME = \Sigma (\text{Forecast} - \text{Actual}) / n$$

- **Mean Absolute Error (MAE):** The average absolute difference between forecasts and actuals.

$$MAE = \Sigma |\text{Forecast} - \text{Actual}| / n$$

- **Root Mean Square Error (RMSE):** The square of the differences between each actual and predicted value, averaging these squared differences, and then taking the square root of the result.

$$RMSE = \sqrt{[\Sigma (\text{Forecast} - \text{Actual})^2 / n]}$$



ACCURACY MEASURES

- **Mean Percentage Error (MPE):** Calculates the average error as a percentage of actual values.

$$\text{MPE} = (\sum (\text{Forecast} - \text{Actual}) / \text{Actual}) / n * 100$$

- **Mean Absolute Percentage Error (MAPE):** Similar to MAE but expressed as a percentage.

$$\text{MAPE} = (\sum |(\text{Forecast} - \text{Actual}) / \text{Actual}| / n) * 100$$

Generally, lower values of RMSE, MAE, and MAPE desirable, indicating more accurate forecasts. Additionally, we consider the sign of ME, MPE.

```
> #Diagnostic checking
> accuracy(c(83.27261,83.27514,83.27753,83.27893),c(83.28,83.26,83.27,83.22))
              ME          RMSE          MAE          MPE          MAPE
Test set -0.0185525 0.03087587 0.0222475 -0.02229138 0.02672821
```



LJUNG BOX TEST

The Ljung-Box test is a statistical test that checks if **autocorrelation** exists in a time series.

The **test statistic** for Ljung box test is :

$$Q_* = n(n+2) \left(\frac{\hat{r}_1^2}{n-1} + \frac{\hat{r}_2^2}{n-2} + \dots + \frac{\hat{r}_K^2}{n-K} \right)$$

where K is selected large enough such that the Ψ -weights are negligible for $j > K$. We would reject the model if the observed value of Q_* exceeded an appropriate critical value in a chi-square distribution with $K - p - q$ degrees of freedom.



LJUNG BOX TEST

For testing, we set the hypothesis as follows:

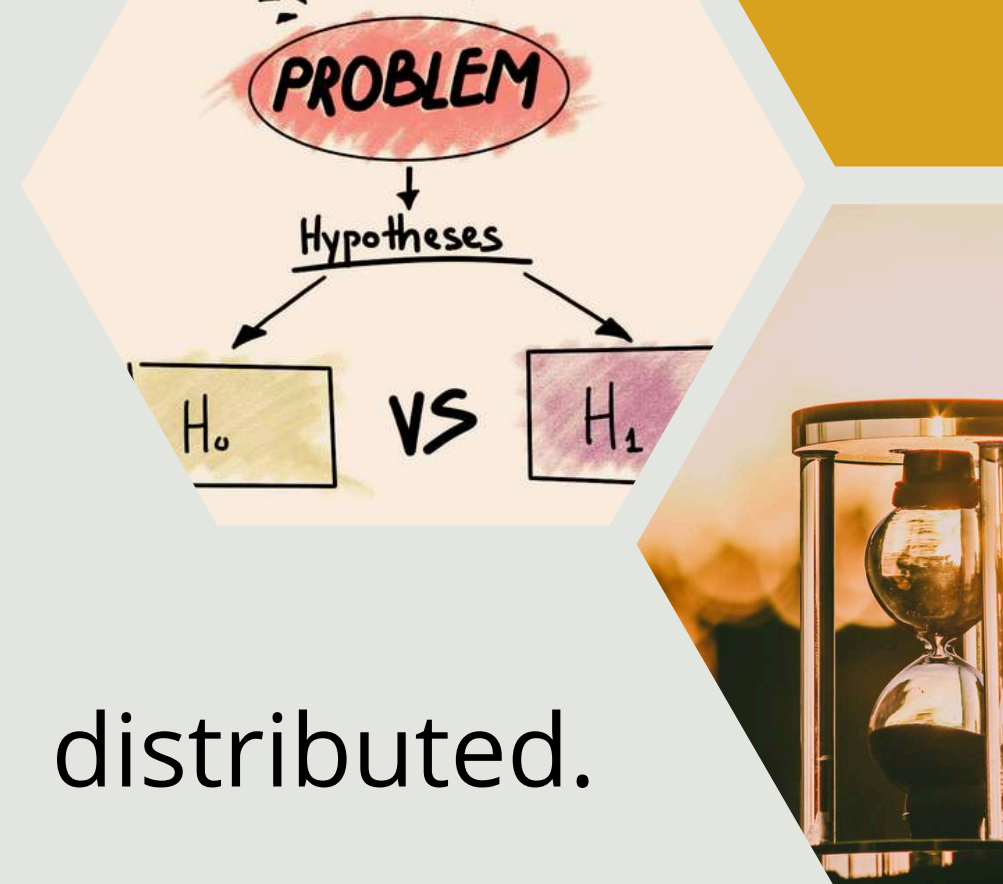
Null hypothesis, H_0 : The residuals are independently distributed. i.e., there is no autocorrelation.

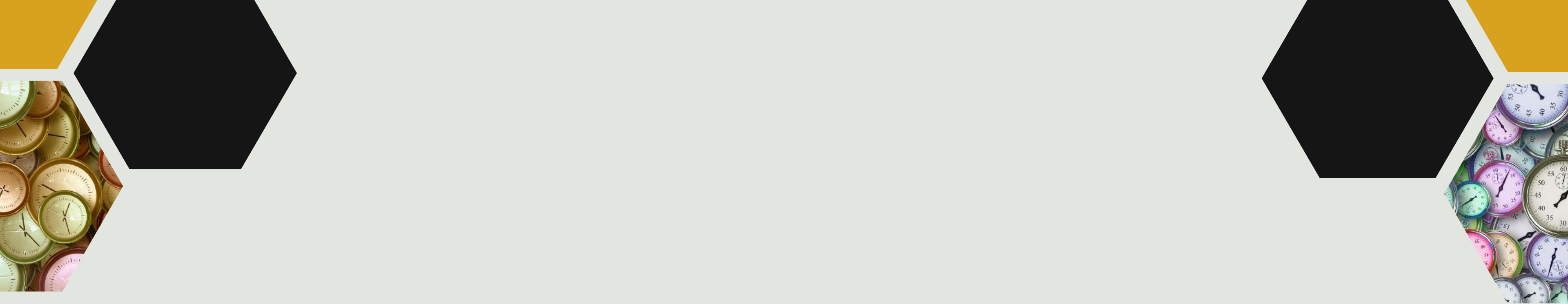
Alternative hypothesis H_1 : The residuals are not independently distributed. i.e., there is autocorrelation.

Box-Ljung test

```
data: residuals  
X-squared = 0.0031114, df = 1, p-value = 0.9555
```

Since p value for the Ljung box test is 0.9555 which is greater than 0.05, hence we may accept null hypothesis at 5% level of significance. Which implies that there is not autocorrelation in the data .





FACTORS

IMPORT - EXPORT

- **Trade Balance:** When a country exports more than it imports (trade surplus), its currency can appreciate due to increased demand. In a trade deficit (more imports than exports), the currency may depreciate.
- **Supply and Demand for Currency:** Exports boost demand for the domestic currency, strengthening the exchange rate. Imports increase the supply of domestic currency, potentially causing depreciation.
- **Price Impact:** Exchange rates affect imported and exported goods' prices. A strong currency makes imports cheaper but can make exports more expensive, influencing trade balances. A weak currency can boost exports and narrow trade deficits.



INFLATION

- **Purchasing Power Erosion:** High inflation reduces the purchasing power of a country's currency, making goods and services more expensive. This decreases the currency's value relative to other currencies.
- **Interest Rate Adjustments:** To control inflation, central banks may raise interest rates. Higher interest rates can attract foreign capital, increasing demand for the country's currency and potentially leading to currency appreciation.
- **Exchange Rate Impact:** The combination of eroding purchasing power due to inflation and higher interest rates can influence exchange rates. It often results in a stronger exchange rate, indicating currency appreciation.



INTEREST RATES

- Interest rates and exchange rates are all highly correlated.
- Higher interest rates in a country can attract foreign capital, leading to an appreciation of the country's currency. Conversely, lower interest rates can lead to depreciation.
- Time series analysis of interest rates and exchange rates can help identify historical trends and relationships, providing valuable insights for understanding and predicting currency movements



GOLD RATE

- Gold is often considered a safe-haven asset in times of economic uncertainty or geopolitical instability. When these factors lead to increased demand for gold, it can result in a stronger exchange rate for the country that exports gold.
- Gold is seen as a hedge against inflation. In periods of high inflation, the demand for gold may rise, leading to an increase in the exchange rate of the country that produces or exports gold.
- Gold is a valuable commodity, and its price is influenced by global supply and demand dynamics.



BREXIT

UK's decision to leave the EU in 2016 had a profound impact on currency markets, particularly the GBP, which experienced a sharp depreciation in value against major currencies during the referendum, including the USD and the Euro. This depreciation was due to uncertainty about UK's economic and political future. Apart from having implications for various economic sectors including imports, exports, and tourism, it had an impact on Gold investment. As GBP depreciated, gold prices surged in GBP terms.



POUND TO USD



Data from Refinitiv · Disclaimer

POUND TO USD

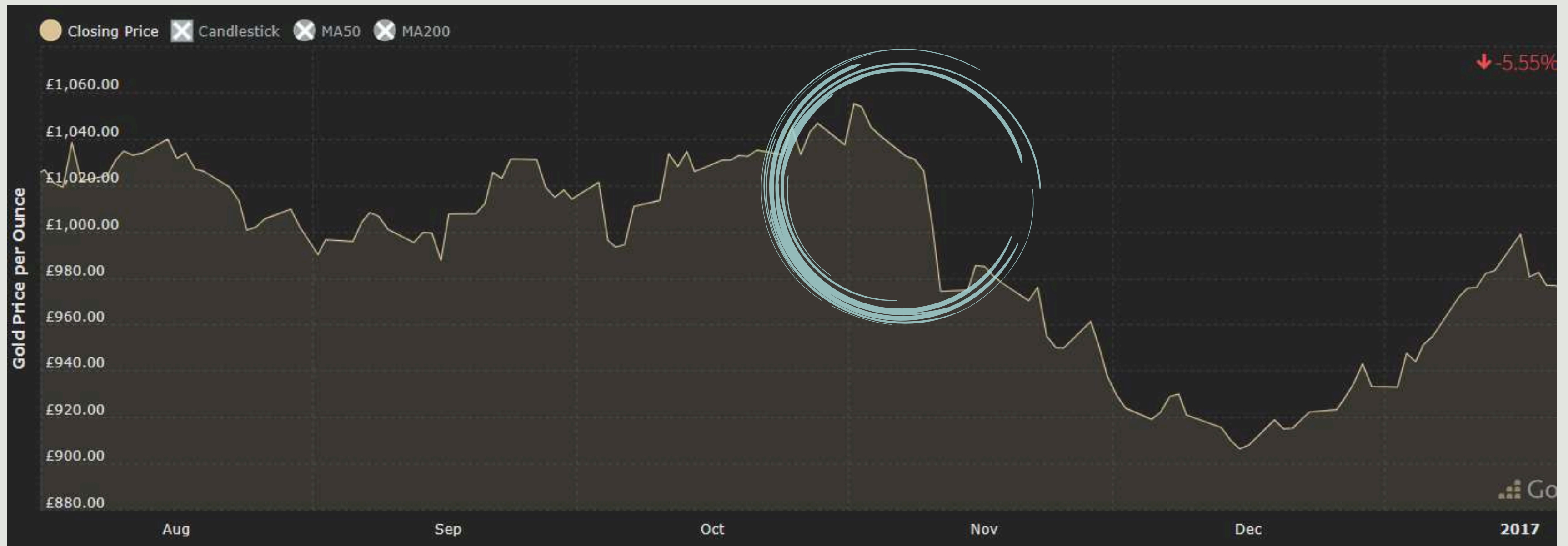


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POUND TO USD



POUND TO USD



COINTEGRATION TEST

- Cointegration is a technique used to find a possible correlation between time series processes in the long term.
- Nobel laureates Robert Engle and Clive Granger introduced the concept of cointegration in 1987.
- The most popular cointegration tests include Engle-Granger, the Johansen Test, and the Phillips-Ouliaris test.

ENGLE - GRANGER TEST

The Engle-Granger Two-Step method starts by creating residuals based on the static regression and then testing the residuals for the presence of unit roots.

It uses the Augmented Dickey-Fuller Test (ADF) or other tests to test for stationarity units in time series. If the time series is cointegrated, the Engle-Granger method will show the stationarity of the residuals.

TEST - RESULT

Null Hypothesis (H0): There is no cointegration between the time series, and they do not have a long-term relationship.

Alternate Hypothesis (H1): There is cointegration between the time series. There is a long-term relationship between the time series.

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 260 on 5630 degrees of freedom
Multiple R-squared:  0.5138,    Adjusted R-squared:  0.5137
F-statistic: 2975 on 2 and 5630 DF, p-value: < 2.2e-16
```

Since, p-value comes out to be less than 0.05 (level of significance), we reject H0 at 5% level of significance. Hence, there is a long-term relationship between the Gold rate and exchange rate.

GRANGER-CAUSALITY TEST

Granger Causality test is used to determine whether one time series can predict another.

If variable X granger causes Y , then past values of X should contain info that helps predict future values of Y

Based on the notion of predictive causality rather than the traditional notion of causality in the sense of one variable causing another

PROCEDURE

For testing, we set the hypothesis as follows:

Null hypothesis, H0: Variable X **does not** Granger-cause variable Y

Alternative hypothesis H1: Variable X Granger-causes variable Y

Using R, we type the following code to get the test result:

```
Granger causality test
```

```
Model 1: data$Exchange.rate.FD ~ Lags(data$Exchange.rate.FD, 1:1) + Lags(data$Gold.rate.FD, 1:1)
```

```
Model 2: data$Exchange.rate.FD ~ Lags(data$Exchange.rate.FD, 1:1)
```

	Res.Df	Df	F	Pr(>F)
1	5630			
2	5631	-1	42.054	9.634e-11 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since, p-value comes out to be less than 0.05(level of significance), we reject H0 at 5% level of significance. Hence, Gold rate granger-causes variable exchange rate.

INFERENCES



Statistically tested and verified that:

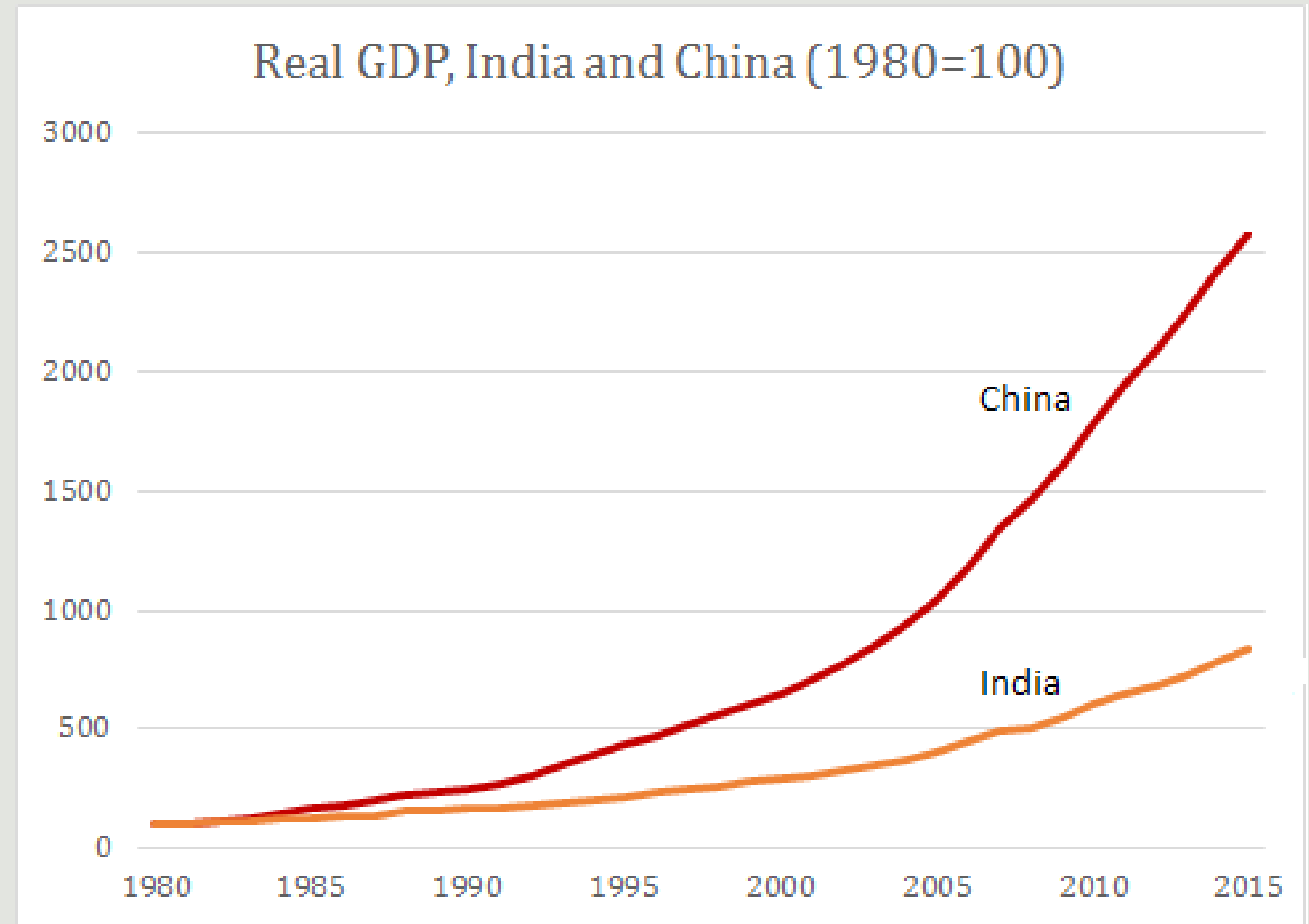
- 1) The exchange rate of rupee to USD has shown an **overall increasing trend** over the past 3 decades.
- 2) Current exchange rate of rupee to dollar **depends on the previous** values of the exchange rate
- 3) **Predicted** the future values of the exchange rate, which helps in **international investment decisions**
- 4) The **gold rate** in India (i.e., price of gold per unit of weight) significantly **influences** the exchange rate in India



HOW EXCHANGE RATE IS INFLUENCING THE WORLD AROUND US?

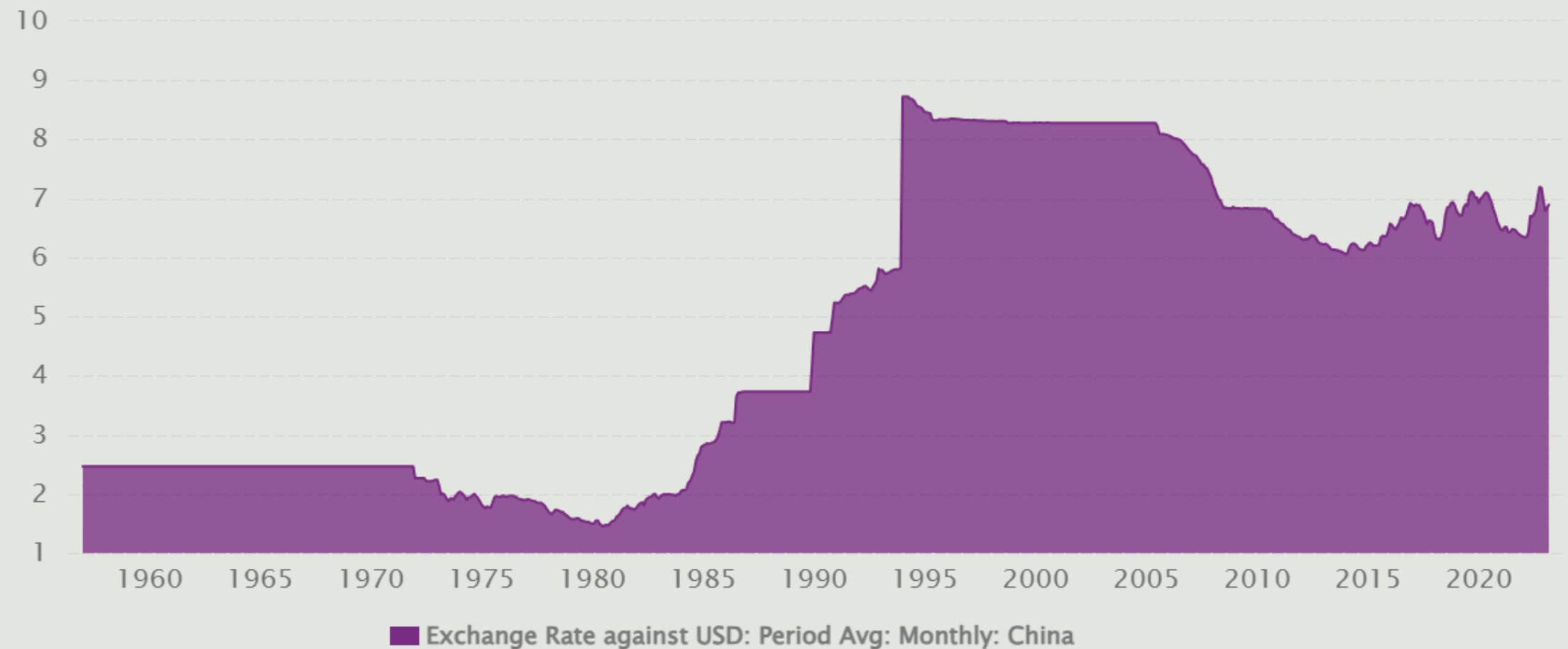
From Fixed Peg to Managed Float: The Transition in China's Exchange Rate System (1957-2023)

**CHINA'S GROWTH
STORY:
Exchange Rate has a
big role to play here**



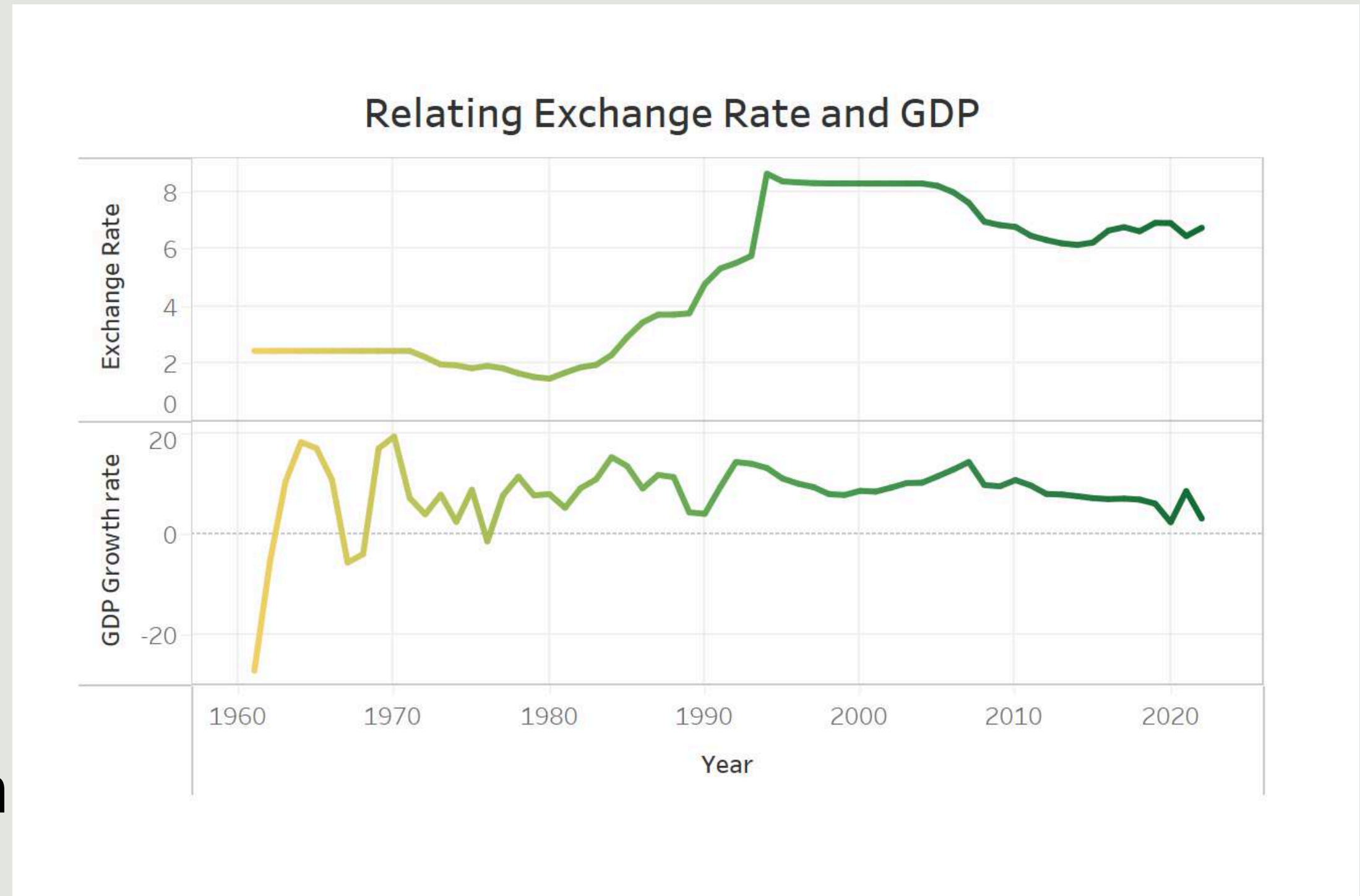
Government-aided development

Transitioned from a fixed economy(uptill 1978) to one with dual exchange rate(1978-1994) to that of a basket determined rate(2005)



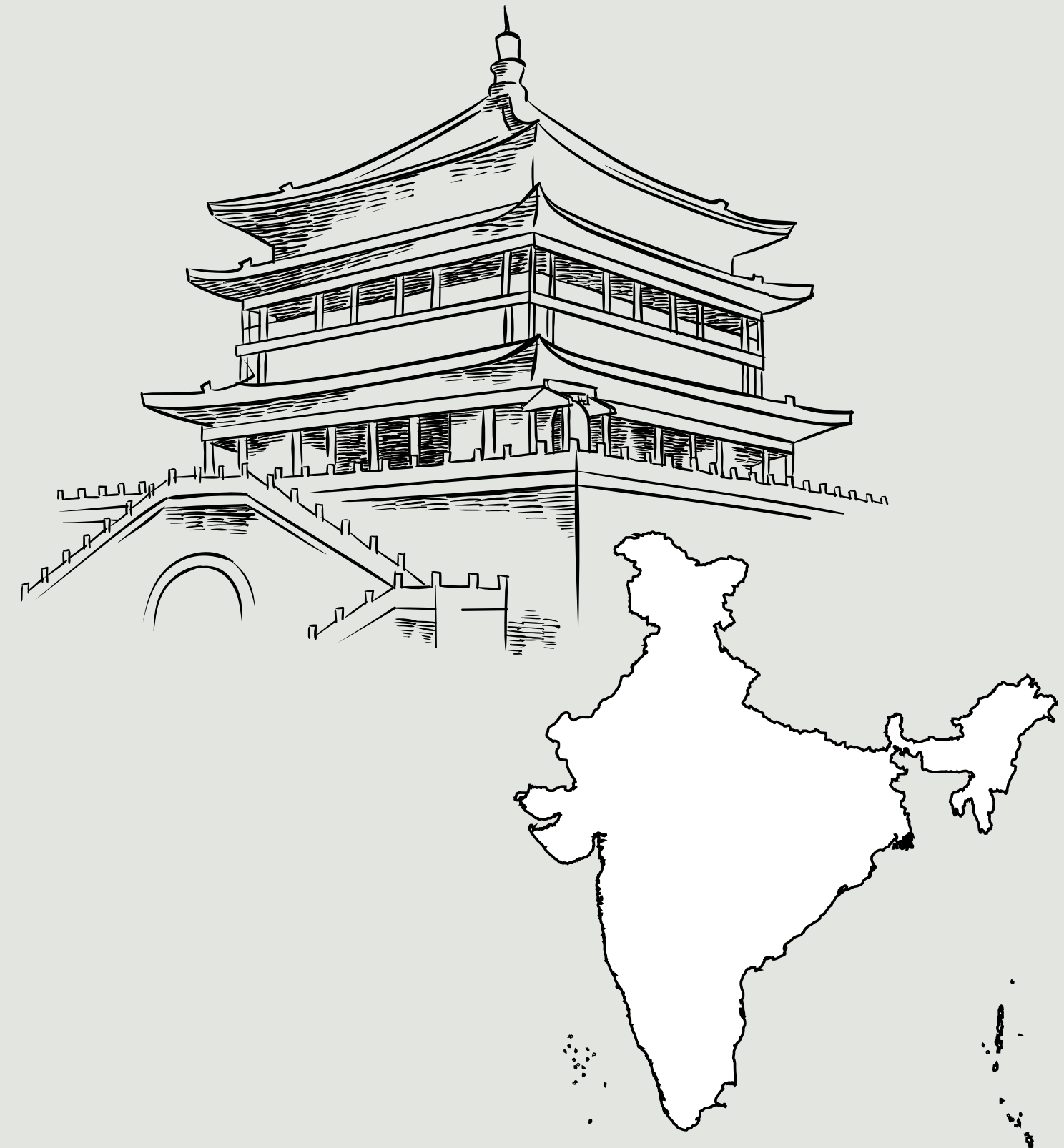
What drove the growth?

- Export oriented growth
Led to manufacturing sector boom.
- Invested in human capital and research
- Dual exchange rate-boosted fdi
- Confidence in Chinese Yuan



LEARNINGS FOR INDIA

- **Focus on internationalization of rupee**(enabling commercial borrowings in rupees).
- **Gradual and government led reforms**(taking stakeholders in confidence)
- **Infrastructure and human capital** development



REFERENCES

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- The R Book by Michael J.Crawley, second edition
- The Analysis of Time Series An Introduction with R, Seventh Edition Chris Chatfield, Haipeng Xing
- www.ijemr.net (International Journal of Engineering and Management Research)



THANK YOU !!